

FA 9 - The Chi-Square Test and Method of Least Squares

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Github Repository: <https://github.com/amyahespanol/Amyah-Espanol-Statistical-Theory-APM1111>

Problem 1

Three coins were tossed a total of 240 times, and each time the number of heads turning up was observed. The results are shown in Table 1, together with the results expected under the hypothesis that the coins are fair. Test this hypothesis at a significance level of $\alpha = 0.05$.

Hypothesis

- H_0 : The coins are fair (binomial distribution where $p = 0.5$).
- H_a : The coins are not fair.

Compute

The expected frequencies for $n = 3$ tosses and $p = 0.5$ are calculated using $E_i = N \times P(X = i)$. $P(X = i) = \binom{3}{i}(0.5)^i(0.5)^{3-i}$.

```
observed <- c(24, 108, 95, 23)
expected <- c(30, 90, 90, 30)

# Chi-square -> sum((O-E)^2 / E)
chi_sq_val <- sum((observed - expected)^2 / expected)
p_val <- 1 - pchisq(chi_sq_val, df = 3)

data.frame(Heads = 0:3, Observed = observed, Expected = expected) |>
  kable(caption = "Coin Tossing Frequencies")
```

Table 1: Coin Tossing Frequencies

Heads	Observed	Expected
0	24	30
1	108	90
2	95	90
3	23	30

Result

- $\chi^2 = 6.7111$
- $df = 3$
- $p\text{-value} = 0.0817$

Final Answer

Since the $p\text{-value}(0.0817) > 0.05$, we can't deny the null hypothesis. There is no significant change to point to the coins being unfair.

Problem 2

Table 2 shows the numbers of students in each of two classes, A and B, who passed and failed an examination given to both groups. Using the (a) $\alpha = 0.05$ and (b) $\alpha = 0.01$ significance levels, test the hypothesis that there is no difference between the two classes. Work the problem with and without Yates' correction.

Hypothesis

- H_0 : There is no difference between classes (independent).
- H_a : There is a significant difference between classes.

Compute

```
exam_table <- matrix(c(72, 64, 17, 23), nrow = 2,
                     dimnames = list(Class = c("A", "B"), Result = c("Passed", "Failed")))
# Without Yates' Correction
test_no_yates <- chisq.test(exam_table, correct = FALSE)
```

```
# With Yates' Correction
test_yates <- chisq.test(exam_table, correct = TRUE)
```

Result

- Without Yates: $\chi^2 = 1.348$, $p = 0.2456$
- With Yates: $\chi^2 = 0.9627$, $p = 0.3265$

Final Answer

Using both $\alpha = 0.05$ and $\alpha = 0.01$, the p -values for both tests are greater than the significance levels. We can't reject the null hypothesis meaning there is no significant difference between Class A and Class B.

Problem 3

Table 3 shows the relationship between hair and eye color of a sample of 200 students.

- Find the coefficient of contingency without and with Yates' correction.
- Compare the result of part (a) with the maximum coefficient of contingency.

(a) Coefficient of Contingency (C)

$$C = \sqrt{\frac{\chi^2}{\chi^2 + N}}$$

```
hair_eye <- matrix(c(49, 30, 25, 96), nrow = 2)
N <- sum(hair_eye)

# Without Yates
chi_no_yates <- chisq.test(hair_eye, correct = FALSE)$statistic
C_no_yates <- sqrt(chi_no_yates / (chi_no_yates + N))

# With Yates
chi_yates <- chisq.test(hair_eye, correct = TRUE)$statistic
C_yates <- sqrt(chi_yates / (chi_yates + N))
```

- Coefficient of Contingency without Yates: 0.3863
- Coefficient of Contingency with Yates: 0.3779

(b) Maximum Coefficient of Contingency

For a 2×2 table, $C_{max} = \sqrt{\frac{k-1}{k}} = \sqrt{\frac{2-1}{2}} = \sqrt{0.5} \approx 0.7071$.

Comparison: The calculated C (0.3863) is slightly high but a lot less than the maximum possible value of 0.7071, which suggests a semi association between hair and eye color.

Problem 4

The `mtcars` dataset in R contains measurements on 32 automobiles, including miles per gallon (`mpg`) and weight (`wt`, in 1000 lbs).

- (a) Using the full dataset, fit a least-squares regression line with `wt` as the independent variable and `mpg` as the dependent variable.
- (b) Fit a second least-squares regression line with `mpg` as the independent variable and `wt` as the dependent variable.
- (c) Plot the data points and both regression lines on the same set of coordinate axes.
- (d) Interpret the slope of each regression line.

Instruction: Use R to load the dataset: `data(mtcars)`

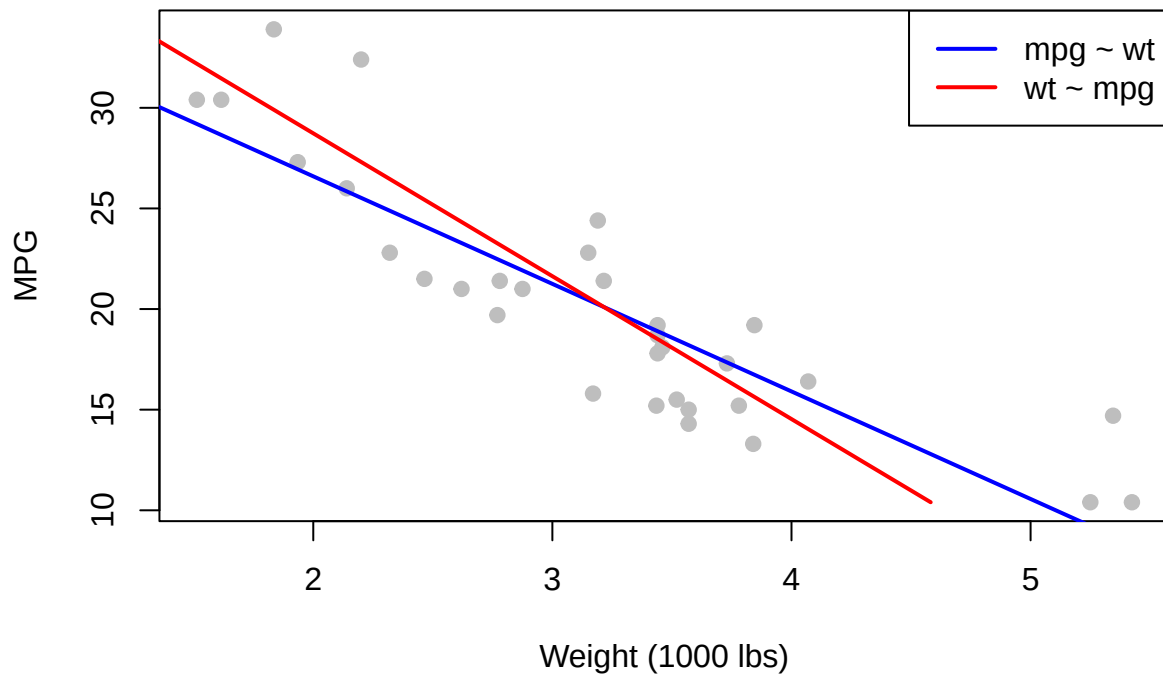
```
data(mtcars)
# (a) wt as independent, mpg as dependent
model1 <- lm(mpg ~ wt, data = mtcars)

# (b) mpg as independent, wt as dependent
model2 <- lm(wt ~ mpg, data = mtcars)
```

(a,b,c) Plot

```
plot(mtcars$wt, mtcars$mpg, pch = 19, col = "gray", xlab = "Weight (1000 lbs)", ylab = "Miles per gallon")
abline(model1, col = "blue", lwd = 2) # part a

# part b & c
mpg_range <- seq(min(mtcars$mpg), max(mtcars$mpg), length.out = 100)
wt_pred <- coef(model2)[1] + coef(model2)[2] * mpg_range
lines(wt_pred, mpg_range, col = "red", lwd = 2)
legend("topright", legend = c("mpg ~ wt", "wt ~ mpg"), col = c("blue", "red"), lwd = 2)
```



(d) Interpret

- Least-squares regression line with wt as the independent variable and mpg as the dependent variable (-5.34): For every increase of 1,000 lbs in weight, the fuel efficiency decreases by approximately 5.34 mpg.
- Least-squares regression line with mpg as the independent variable and wt as the dependent variable (-0.14): For every 1 mpg increase in fuel efficiency, the vehicle weight decreases by approximately 141 lbs.

Problem 5

The cars dataset in R contains measurements of stopping distance (dist, in feet) and speed (speed, in miles per hour) for 50 automobiles.

- Using the full dataset, construct a scatter plot of stopping distance(D) against speed(V).
- Fit a least-squares quadratic model of the form $D = a_0 + a_1V + a_2V^2$.
- On the same graph from part (a), overlay the fitted quadratic regression curve.

- (d) Estimate the stopping distance when $V = 45$ mi/h and $V = 80$ mi/h.
- (e) Fit a simple linear regression model and overlay its fitted line on the same plot. Compare the two models based on their fit.

Instruction: Use R to load the dataset: `data(cars)`

```
data(cars)
# quadratic model
quad_model <- lm(dist ~ speed + I(speed^2), data = cars)

# linear model
lin_model <- lm(dist ~ speed, data = cars)
```

(a,c,e) Scatter plot and Overlay of linear and quadratic

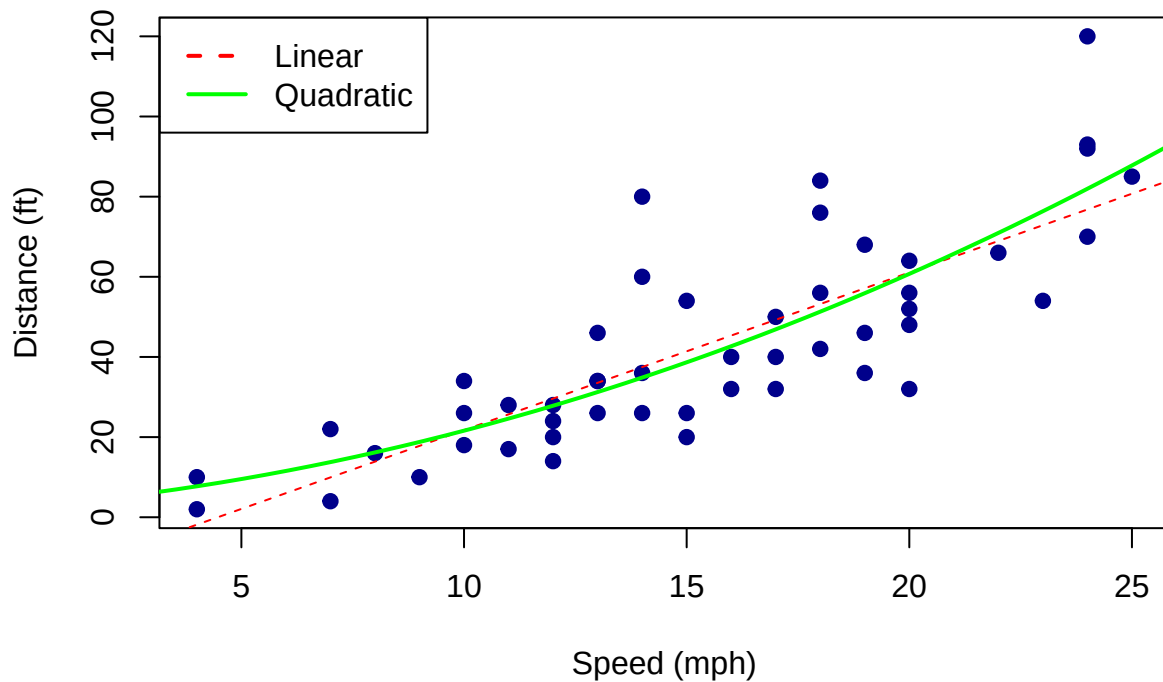
```
plot(cars$speed, cars$dist, pch = 19, col = "darkblue",
     main = "Stopping Distance vs Speed", xlab = "Speed (mph)", ylab = "Distance (ft)")

# Overlay linear
abline(lin_model, col = "red", lty = 2)

# Overlay quadratic
v_vals <- seq(0, 50, length.out = 100)
d_preds <- predict(quad_model, data.frame(speed = v_vals))
lines(v_vals, d_preds, col = "green", lwd = 2)

legend("topleft", legend = c("Linear", "Quadratic"), col = c("red", "green"), lty = c(2,
```

Stopping Distance vs Speed



(d) Estimates

```
new_speeds <- data.frame(speed = c(45, 80))
preds <- predict(quad_model, new_speeds)
```

- Estimated distance at 45 mph: 245.99 ft
- Estimated distance at 80 mph: 715.27 ft

Comparison of the two models

The quadratic model is generally better for this data set. Stopping distance is related to kinetic energy ($KE = \frac{1}{2}mv^2$), which shows a squared variable with velocity. Off of looks, the quadratic curve better shows the upward “bend” in the data as speed increases, whereas the linear model tends to underestimate distances at very high speeds.